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Q: What is micro programmed control unit?

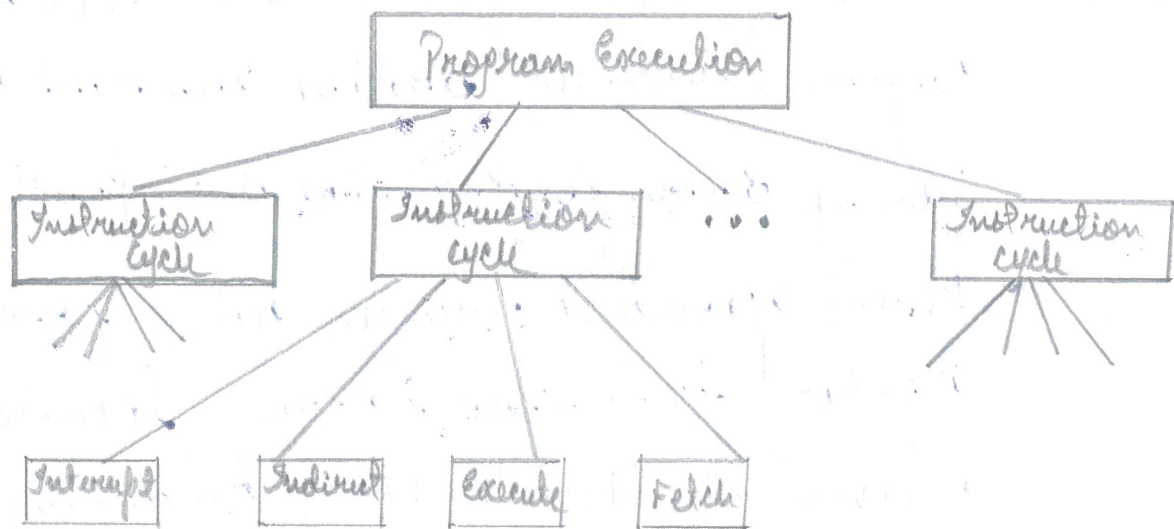
Ans: A microprogramming control unit is component of computer's CPU that uses a set of microinstruction to manage the execution of machine-level instruction.

Q: What are the basic functionalities of micro operations?

Ans: • These are the simplest actions a CPU can perform, such as adding numbers or moving data, fundamental to its functions

- Micro operations involve using processor registers - small, fast storage location within the CPU to hold data temporarily for quick calculation.
- Micro signifies low-level tasks, like transferring a bit or incrementing a value which collectively execute complex instruction

- Macro programmed instructions are executed sequentially, enabling the CPU to complete task systematically.
- The instruction cycle is the process where the CPU retrieves & process an instruction.
- The cycle includes sub-cycles like: fetching from memory, addressing etc.
- Each sub cycle involve specific micro operation such as loading data into registers, essential to the overall instruction cycle.



Every program that is executed is divided into ~~system~~ many instruction cycles, each cycles responsible for a specific action.

1) Fetch cycle

2) Indirect cycle

3) Execute cycle

4) Interrupt cycle

Fetch Cycle:-

At the beginning of each instruction cycle, the fetch phase retrieves the next instruction from memory using the address stored in the program counter (PC). This instruction is then held temporarily in the CPU for subsequent decoding & execution.

4 registers are:-

1) Memory Address Register (MAR):

- It is connected to address bus i.e. responsible for carrying the memory address from the CPU to other components.
- It also specifies the address for read or write.

2) Memory Buffer Register (MBR):

- It is connected to data bus i.e. responsible for transferring actual data betⁿ the CPU, memory, & other components of the system.
- It holds data to write into memory or the data i.e. retrieved from the memory it is temporarily held in the MBR before being used in further ~~op~~ operation.

3) Program Counter (PC):

- Register that holds the memory address of the

- Register that temporarily stores the instruction fetched from memory during the instruction cycle for processing.

MAR	
MBR	
PC	000000001100100
IR	
AC	

(a) Beginning

MAR	000000001100100
MBR	000100000100000
PC	000000001100101
IR	
AC	

(c) After Second Step

MAR	000000001100100
MBR	0000000001100100
PC	0000000001100100
IR	
AC	

(b) After first step

MAR	000000001100100
MBR	000100000100000
PC	000000001100101
IR	000100000100000
AC	

(d) After Third Step

The execution of the fetch cycle is divided into 3 times periods namely t_1, t_2, t_3

t_1 Move the address from the program Counter (PC) to the memory Address Register (MAR)

t_2

- Place the address in the MAR onto the address bus
- Control unit issues a READ command on the control bus
- The instruction from memory at the specified address is placed on the data bus & copied into the Memory Buffer Register (MBR)

t_3

- Transfer the instruction from the MBR to the Instruction Register (IR)
- This frees the MBR for any potential indirect operations.

$t_1: \text{MAR} \leftarrow (\text{PC})$
 $t_2: \text{MBR} \leftarrow \text{Memory}$
 $\text{PC} \leftarrow (\text{PC})$
 $t_3: \text{IR} \leftarrow (\text{MBR})$

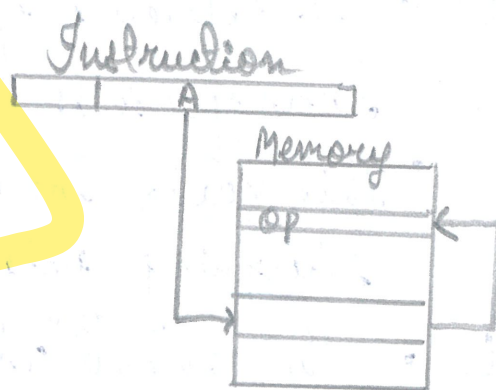
Indirect Cycle: In the indirect cycle with indirect addressing mode, the memory address provided by the instruction is not the actual location of the data, but a pointer to another address where the data is stored. During this cycle, the CPU fetches the indirect address from memory, then uses it to retrieve the actual data. This extra step allows for more flexible memory management & enable accessing a larger range of memory location.

The execution of the indirect cycle is divided into 3 time periods namely t_1, t_2, t_3

$t_1: \text{MAR} \leftarrow (\text{IR}(\text{Address}))$

$t_2: \text{MBR} \leftarrow \text{Memory}$

$t_3: \text{IR}(\text{Address}) \leftarrow (\text{MBR}(\text{Address}))$



(c) Indirect

t_1 The address field of instruction in the instruction Register (IR) is transferred to the memory Address Register (MAR).

⇒ The address in the MAR is used to fetch the actual operand address from memory, which is stored in the memory buffer register (MBR).

- t₃ • Transfer the instruction from the MBR to the instruction Register (IR).
- This frees the MBR for any potential indirect operation.

Interrupt Cycle: The interrupt cycle saves the CPU current state & transfers control to an interrupt handler to address a highly-priority event. After handling, the interrupt the CPU resumes the interrupted task. The execution of the interrupt cycle is divided into 3 time periods namely t_1, t_2, t_3 .

- t₁ • The contents of the program Counter (PC) are transferred to the memory buffer Register (MBR).
- This saves the current instruction address for returning after the interrupt.
- t₂ • The memory address register (MAR) is loaded with address where the PC's content will be saved.
- The PC is then loaded with the address of the start of the interrupt processing routine.
- t₃ • The PC is then loaded with address of the interrupt processing routine.

$t_1: \text{MBR} \leftarrow (\text{PC})$

$t_2: \text{MAR} \leftarrow \text{Same address}$

$\text{PC} \leftarrow \text{Routine address}$

$t_3: \text{Memory} \leftarrow (\text{MBR})$

Execute Cycle:

- **Operation Execution:** The ALU performs the specified operation on the operand.
- **Operand fetch:** Operand are fetched from memory or register if needed.
- **Result Storage:** The result is stored in a designated register or memory location.
- **Factors Affecting Performance:** Instruction complexity, operand availability, ALU design, and pipeline design influence execution speed.

The control unit examines the opcode & generates a sequence of micro-operation based on the value of the opcode.

Example

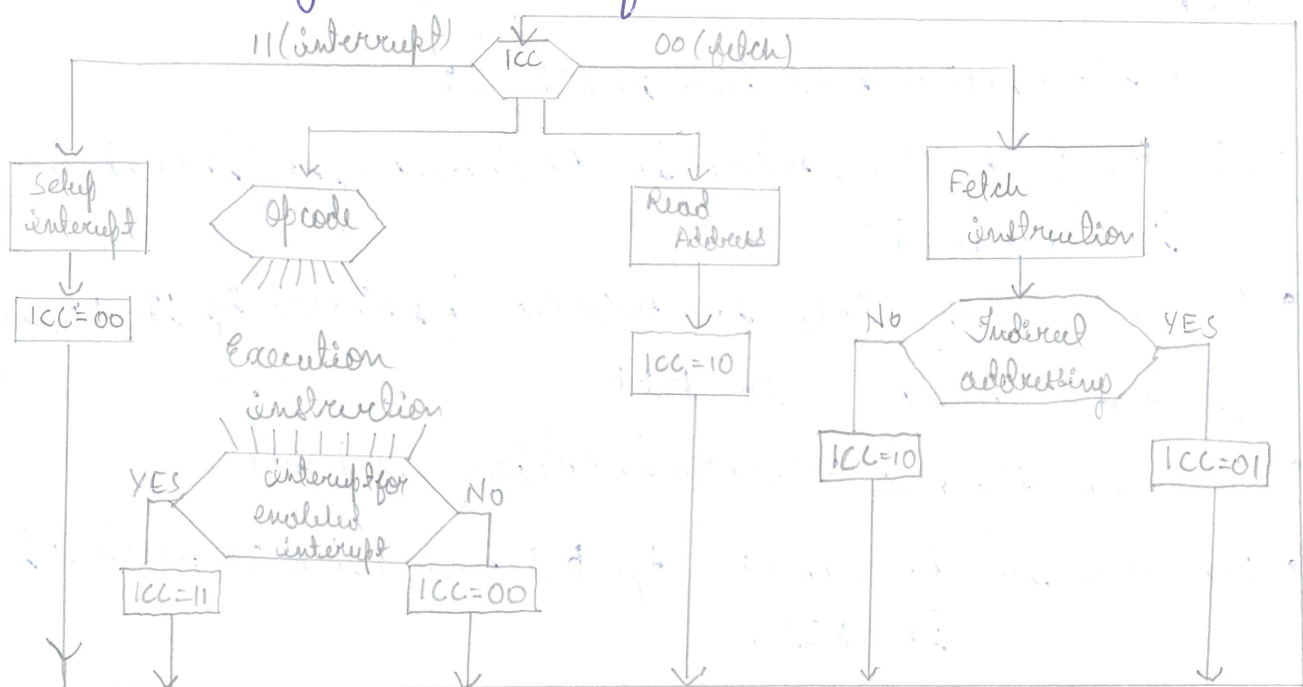
A simplified add instruction:

- **ADD R1, x** (which adds the contents of the location x to register R1)
- In the first step the address portion of the IR is loaded into the MAR
- Then the referred memory location is Read
- Finally the contents of R1 and MBR are added by the ALU.

extract the register reference from the IR & perhaps to stage the ALU inputs or outputs in some intermediate register.

Rules for Micro operations groups:

- Proper sequence must be followed (I mean $\leftarrow$) MAR
[PC] must precede MBR [memory]
- Conflicts must be avoided
- Must not read & write same register at same time
MBR [memory] & IR [MBR] must not be in same cycle
- One of the micro-operations involves an addition
- To avoid duplication of circuitry, this addition could be performed by the ALU.
- The use of the ALU may involve additional micro-operations, depending on the functionality of the ALU & the organization of the processor.



Instruction Cycle Code (ICC):

- A new 2-bit register called the Instruction Cycle Code (ICC) is introduced to indicate the current phase of the instruction cycle:
 - 00: fetch
 - 01: indirect
 - 10: execute
 - 11: interrupt
- At the end of each cycle, the ICC is updated to reflect the next state.

flow of Cycles:

- The indirect cycle is always followed by the execute cycle.
- The interrupt cycle is always followed by the fetch cycle.
- For both the fetch & execute cycles, the next phase depends on the current system state.

Micro operations:

- Each phase involves specific sequence of Micro operation that define how the processor interacts with its registers, ALU, & memory.

Control Unit Role:

- The control unit orchestrates the execution of these micro operations based on the state of the ICC & the instruction sequence.
- The control flow is more complex in actual processor but follows a similar structure emphasizing that the control flow is essentially a sequence of micro-

Comparison of Hardware & Microprogrammed Control units:

Feature: Implementation

- **Hardware Control unit:** The control logic is implemented using fixed logic circuit such as AND, OR & NOT gates. This makes the control unit inflexible & difficult to modify.
- **Microprogrammed control unit:** The control logic is stored in a control memory as a sequence of microinstructions. This allows for greater flexibility.
- **Differences:** The hardware control unit has a fixed structure while the microprogrammed control unit can be reprogrammed by changing the microinstruction.

Feature: Control word generation

- **Hardware control unit:** The control word is generated directly from combinational logic circuit. This means that the control word is determined by the current state of the machine & the input signal.
- **Microprogrammed control unit:** The control word is stored in the control memory as a microinstruction. This allows for more complex control sequence & easier modification of the control logic.
- **Differences:** The hardware control unit may have limitations in the complexity of control sequences, while the microprogrammed control unit can handle complex logic.

Feature: Control Memory

- **Hardwired Control unit:** No control memory is used. The control logic is implemented directly in hardware.
- **Microprogrammed Control unit:** A control memory is used to store the microinstruction that define the control logic. This allows for greater flexibility & ease of modification.
- **Differences:** The hardwired control unit does not require additional hardware for control while the microprogrammed control unit requires a control memory.

Feature: Data path

- **Hardwired Control unit:** The data path is more flexible and less adaptable. This means that the data flow through the machine is predetermined and cannot be easily changed.
- **Microprogrammed Control unit:** The data path is more flexible & can be adapted easily. This allows for greater versatility in the machine's functionality.
- **Differences:** The hardwired control unit has a limited data path configuration while the micro-programmed control unit can support different data path configurations.

- **Hardwired control unit:** Generally faster due to its direct implementation.
- **Microprogrammed control unit:** Generally slower due to the need to fetch microinstruction from memory.
- **Differences:** The hardwired control unit has a faster execution time due to its direct hardware implementation.

Feature: flexibility

- **Hardwired control unit:** less flexible, hard to modify.
- **Microprogrammed control unit:** More flexible, easy to modify.
- **Differences:** The hardwired control unit may be limited in its functionality, while the microprogrammed control unit can be adapted to support different operations.

Feature: complexity

- **Hardwired control unit:** increased with instruction set complexity.
- **Microprogrammed control unit:** can manage complexity through organized microinstruction.
- **Differences:** The hardwired control unit becomes more complex as the instruction set grows, while the microprogrammed control unit can handle complex instruction sets through well organized microinstructions.

Feature: Cost

- Hardwired control unit: Typically cheaper in simpler designs.
- Microprogrammed control unit: Potentially more expensive due to memory.
- Differences: The microprogrammed control unit may be more expensive due to the cost of the control memory.

Feature: Design Change

- Hardwired control unit: Requires hardware redesign.
- Microprogrammed control unit: Only requires changes to microprogram.
- Differences: The hardwired control unit requires physical modifications to the hardware to change its functionality.

Feature: Micro-Operation Control

- Hardwired control unit: Directly from combinational logic.
- Microprogrammed control unit: From stored microinstructions.
- Differences: The hardwired control unit generates micro-operations directly from its logic circuitry.

- Hardwired control unit: 11010000 (fixed for an operation)
- Microprogrammed control unit: 11000000, 00010000, etc.
(varies per microinstruction)
- Differences: The hardwired control unit has a fixed control word for each operation.

Feature: Execution flexibility

- Hardwired control unit: limited; hard to modify control logic.
- Microprogrammed control unit: H/w can easily change control words by updating microinstruction
- Differences: The hardwired control unit has limited flexibility in modifying its control logic.

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